গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
সিলেট-৩১১৪, বাংলাদেশ।
যোগাযোগঃ +৮৮ ০১৩১৮ ২১০ ২৮৬/২৮৭/২৮৮/২৮৯/২৯০
Website: www.crtc-cee-sust.com

Center for Research, Testing and Consultancy
Department of Civil and Environmental Engineering
Shahjalal University of Science and Technology
Sylhet-3114, Bangladesh.
Contact: +88 01318 210 286/ 287/ 288/ 289/ 290
Email: crtc-cee@sust.edu

Ref. no: CRTC/CEE/20220450/S- 82

SUB-SOIL INVESTIGATION REPORT FOR

PROPOSED CONSTRUCTION OF 02 STORIED BUILDING OF SUSTAINABLE FORESTS & LIVELTHOODS (SUFAL) PROFECT-01

AT CHUNARUGHAT STAFF BARRAK, CHUNARUGHAT, HOBIGONJ

CLIENT

EXECUTIVE ENGINEER, PWD, HOBIGONJ

CONDUCTED BY
CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

Sample Collected by
M/S BSHAL CONSTRUCTION
Chunarughat Beat office staff quarter,
Chunarughat, Hobigonj
Mobile: 01711-300724

Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-701ft	14
02	Disturbed sample	02	0-70 ft	14
03	Disturbed sample	03	0-70 ft	14
04	Disturbed sample	04	0-70 ft	14

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	**Depth**	**Design N**	**Clay**						**Sand**
			Continuous			**Square(Isolated)**			
	(ft)	**-**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	
BH01	**2.5-5.0**	**7**	0.789	0.830	0.904	0.932	1.006	1.095	0.850
	5.0-10	**12**	1.352	1.422	1.550	1.597	1.725	1.876	1.457
BH02	**2.5-5.0**	**6**	0.742	0.781	0.851	0.877	0.947	1.030	0.800
	5.0-10	**11**	1.232	1.296	1.413	1.455	1.572	1.710	1.328
BH03	**2.5-5.0**	**8**	0.882	0.927	1.011	1.041	1.125	1.224	0.950
	5.0-10	**12**	1.392	1.464	1.596	1.644	1.776	1.932	1.500
BH04	**2.5-5.0**	**8**	0.882	0.927	1.011	1.041	1.125	1.224	0.950
	5.0-10	**12**	1.352	1.422	1.550	1.597	1.725	1.876	1.457

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole			
ft	BH01	BH02	BH03	BH04
5	0.049	0.049	0.049	0.049
10	0.049	0.049	0.049	0.049
15	0.112	0.104	0.128	0.128
20	0.144	0.128	0.136	0.128
25	0.128	0.136	0.144	0.136
30	0.156	0.118	0.096	0.096
35	0.138	0.096	0.156	0.096
40	0.196	0.118	0.118	0.118
45	0.138	0.156	0.156	0.096
50	0.156	0.174	0.174	0.156
55	0.468	0.508	0.468	0.508
60	0.669	0.669	0.669	0.669
65	0.669	0.669	0.669	0.669
70	0.669	0.669	0.709	0.669

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole			
ft	BH01	BH02	BH03	BH04
5	0.450	0.450	0.450	0.450
10	0.450	0.450	0.450	0.450
15	22.400	20.800	25.600	25.600
20	28.800	25.600	27.200	25.600
25	25.600	27.200	28.800	27.200
30	1.575	1.125	0.900	0.900
35	1.350	0.900	1.575	0.900
40	2.025	1.125	1.125	1.125
45	1.350	1.575	1.575	0.900
50	1.575	1.800	1.800	1.575
55	7.875	8.550	7.875	8.550
60	11.250	11.250	11.250	11.250
65	11.250	11.250	11.250	11.250
70	11.250	11.250	11.925	11.250

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole			
ft	BH01	BH02	BH03	BH04
5	0.032	0.032	0.032	0.032
10	0.032	0.032	0.032	0.032
15	0.073	0.068	0.083	0.083
20	0.094	0.083	0.088	0.083
25	0.083	0.088	0.094	0.088
30	0.101	0.077	0.062	0.062
35	0.090	0.062	0.101	0.062
40	0.127	0.077	0.077	0.077
45	0.090	0.101	0.101	0.062
50	0.101	0.113	0.113	0.101
55	0.304	0.330	0.304	0.330
60	0.435	0.435	0.435	0.435
65	0.435	0.435	0.435	0.435
70	0.435	0.435	0.461	0.435

Table 11: The allowable end bearing for Bored piles in Tsf (FS=3.0)

Depth	Bore Hole			
ft	BH01	BH02	BH03	BH04
5	0.203	0.203	0.203	0.203
10	0.203	0.203	0.203	0.203
15	10.080	9.360	11.520	11.520
20	12.960	11.520	12.240	11.520
25	11.520	12.240	12.960	12.240
30	0.709	0.506	0.405	0.405
35	0.608	0.405	0.709	0.405
40	0.911	0.506	0.506	0.506
45	0.608	0.709	0.709	0.405
50	0.709	0.810	0.810	0.709
55	3.544	3.848	3.544	3.848
60	5.063	5.063	5.063	5.063
65	5.063	5.063	5.063	5.063
70	5.063	5.063	5.366	5.063

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed three storied residential building.

Soil profile scenario

Overall soil information of the investigated site is regular in between the boreholes, but vertical inconsistency exists in the soil formation.

Gray soft clay with low bearing capacity exists at the upper layer up to 10 ft from EGL, followed by Gray fine sand with silt and clay of medium bearing capacity (15 ft thick layer). Below this layer again there is a layer of Gray soft clay with low bearing capacity. Gray hard clay with good bearing capacity was found at 55 ft depth from EGL. Detailed scenario is provided with the report (please see attached bore-log profiles with SPT values).

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since soft clay with low bearing capacity was found at top layer, proper ground improvement is needed before adopting any shallow foundation. In this regard, the soft soil from the top 10 ft depth is to be excavated first and then a compacted bed of minimum 5 ft thickness should be prepared using brick/stone chips and sand. Sand piling is an alternative option for the proposed 2 storied building. Sand pile dia should be 8 inch and c/c distance of piles should be 18 inch. The bearing capacity for shallow foundation which is given in the **Table 7** is based on existing soil condition. However, client need to reassess the improved bearing capacity of the prepared ground after implementing soil improvement technique in the top layer as mentioned above.

Pile is not an economic option for 2 storied building. However, if piles are considered for the site, size and depth should be selected in the light of bearing capacity given in the report (Tables 8, 9 are for precast piles and Tables 10, 11 are for cast in situ piles).

Safety provisions

It is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **70** ft as client provided.

Report Prepared By	**Test Assisted By**	**Counter Signed By**
Dr. Mushtaq Ahmed Professor Dept. of CEE, SUST	**A.H.M. Humayun Kabir** Senior Technical Officer Dept. of CEE, SUST	**Dr. HMA Mahzuz** Coordinator, CRTC Professor, Dept. of CEE, SUST